H-B-401c
September 25, 1964

SUPERSEDING
Int. Fed. Spec. H-B-00401b (GSA-FSS)
June 25, 1963 and
Fed. Spec. H-B-401a
June 1, 1950

FEDERAL SPECIFICATION

BRUSHES, MUCILAGE AND PASTE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for two types of brushes for general office use (see 6.2).

1.2 Classification. (See 6.1.)

1.2.1 *Types and sizes.* The brushes shall be of the following types and sizes:

Type CR.—With collar and rim (see fig. 1).
Size: 11/16-inch.

Type F.—Flat, without collar (see fig. 2).
Sizes: 1-inch and 3-inch.

Type F brushes shall be furnished in either of the sizes listed above, as specified in the invitation for bids, contract, or order.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-601—Boxes; Wood, Cleated-Plywood.
PPP-B-621—Boxes; Wood, Nailed and Lock Corner.
PPP-B-636—Box; Fiberboard.
PPP-B-665—Boxes; Paperboard, Metal-Stayed (Including Stay Material).
PPP-B-676—Boxes; Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 7510

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Military Specifications:

MIL-L-10547—Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-P-116—Preservation, Methods of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials used in the brushes shall be as specified hereinafter and shall be of the quality normally used for the purpose in good commercial practice.

3.2 Design. The brushes shall be similar in design to those illustrated in figures 1 and 2 (see 6.2).

3.3 Type CR, brushes with collar and rim.

3.3.1 *Bristle.* The bristle shall be pig or hog bristle. All bristle shall be boiled or steamed, straightened, and thoroughly cleaned. The bristle shall show no tendency to separate at the end or to flare out when the brush is dried after washing. (See 4.4.)

3.3.2 *Handle.* The handle shall be of wood, hard rubber, or slow-burning or flameproof plastic. Plastic of cellulose nitrate composition will not be acceptable. Wood handles shall be smoothly finished with varnish, lacquer, or paint. (See 4.4.)

3.3.3 *Ferrule.* The ferrule shall be of corrosion-resistant metal or of metal treated to resist corrosion.

3.3.4 *Rim.* The rim shall fit over the mouth of a bottle approximately 2 inches in diameter, with the brush centered. The rim shall extend downward approximately 3/16 inch.

3.3.5 *Set.* Bristles shall be doubled into the ferrule and flattened at the brush end. The handle shall be inserted into the ferrule so as to butt against the bristle and prevent the bristle tuft from pushing back. The following alternate construction will be acceptable: The bristle shall be inserted in the ferrule and cemented therein. Cementing shall be sufficiently secure to prevent the bristle tuft from pushing back.

Ferrules shall be crimped slightly at the bristle end to give the brush a "paint brush" shape, and shall be tightly swaged to handles.

3.3.6 *Physical requirements.* Type CR brushes shall conform to the requirements specified in table I.

TABLE I.—*Physical requirements for type CR brushes (min.)*

Brush; components		Size 11/16-inch
Ferrule:		
Length	inch	1
Thickness	do	0.012
Bristle:		
Width inside ferrule	do	5/8
Length outside ferrule	do	1
Thickness inside ferrule	do	1/16
Hold in ferrule	do	1/4
Handle:		
Length	inches	1-3/8
Diameter	inch	3/4
Over-all length	inches	3-5/8

3.4 Type F, flat brushes without collar.

3.4.1 *Bristle.* The bristle shall be pig or hog bristle, as specified (see 6.1). All bristle shall be boiled or steamed, straightened, and thoroughly cleaned. The bristle shall show no tendency to separate at the end or to flare out when the brush is dried after washing. (See 4.4.)

3.4.2 *Handle.* The handle shall be wood, hard rubber, or slow-burning or flameproof

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plastic. Plastic of cellulose nitrate composition will not be acceptable. Wood handles shall be smoothly finished with varnish, lacquer, or paint. (See 4.4.)

3.4.3 *Ferrule.* The ferrule shall be of corrosion-resistant metal or of metal treated to resist corrosion.

3.4.4 *Set.* Bristles shall be set in rubber, brushmaker's pitch, epoxy resin compound, or other suitable setting material which will not become brittle or be affected by the material with which the brushes are intended to be used. The setting material used shall be sufficiently secure to prevent removal of the bristles from the brush part of the setting compound without breaking the bristles when tested as specified in 4.3.1. Handle, ferrule, and bristle shall be rigidly secured together.

3.4.5 *Physical requirements.* Type F brushes shall conform to the requirements specified in table II.

TABLE II.—*Physical requirements for type F brushes (min.)*

Brush; components		Sizes	
		1-inch	3-inch
Ferrule:			
Length	inches	1	1-1/4
Thickness	inch	0.012	0.012
Bristle:			
Width inside ferrule	inches	1	3
Length outside ferrule	do	1-1/2	2-1/4
Thickness inside ferrule	inch	5/16	7/16
Hold in ferrule	do	5/16	7/16
Handle:			
Length	inches	4-1/2	5
Over-all length	do	6-3/4	8-3/4

3.5 **Workmanship.** Workmanship shall be in accordance with normal commercial practice. Brushes shall be clean, well made, and free from defects which might impair their serviceability and which affect the appearance of the commodity.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier's responsibility.** Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. (See 4.4 and 6.1.)

4.2 **Sampling and inspection for acceptance.** Sampling and inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the same type and size offered for delivery at one time.

4.2.1 *Inspection of components and materials.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used in the brushes were manufactured, inspected, and tested in accordance with requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, the requirements of this specification shall govern.

4.2.2 *Examinations.*

4.2.2.1 *Lot size.* The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of shipping containers for examination under 4.2.2.3 and in units of brushes for examination under 4.2.2.4.

4.2.2.2 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels for determining the sample

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size and the acceptable quality levels (AQLs), expressed in defects per one hundred units, shall be as indicated in table III.

TABLE III.—*Inspection and acceptable quality levels for examinations*

Reference paragraph	Inspection level	AQL
4.2.2.3	S-2	4.0
4.2.2.4.1	I	4.0
4.2.2.4.2	S-1	2.5

4.2.2.3 *Examination of preparation for delivery.* Examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Examination for defects shall be made in accordance with table IV. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery and selected just prior to the closing operations. Defects of closure shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. (See table III.)

TABLE IV.—*Examination of preparation for delivery*

Examine—	Defects
Marking for shipment—exterior and interior containers	Omitted; incorrect, illegible; of improper size, location, sequence, or method of application.
Materials (containers)	Any component missing; damaged, or otherwise defective affecting serviceability.
Closure of shipping containers	Not as specified.
Contents and weight, exterior and interior, as applicable	Quantity of brushes per package and quantity of packages per shipping container are less than required. Gross or net weight exceeds limitation.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape, or case liners; loose or inadequate sealing, strapping, or stapling. Bulged or distorted containers.
Packaging and packing containers	Not type, class, style, etc., specified, as applicable.

4.2.2.4 *Examination of end items.*

4.2.2.4.1 *For materials, construction, and workmanship.* Examination of end items for these characteristics shall be made in accordance with the classification of defects listed in table V. The sample unit for this examination shall be one brush. (See table III.)

TABLE V.—*Classification of defects (end items)*

Examine—	Defects	Category	
		Major	Minor
Materials	Not as specified ..	X	
Design	Not adequate for purpose of intended use of brush; type CR not with collar and rim and type F not flat; brush not generally similar to figures 1 and 2, respectively	X	
Construction	Rim and set not as specified (type CR brush)	X	
	Bristles of type F brush not set as specified; handle, ferrule, and bristle not rigidly secured together	X	
Dimensions	(See 4.2.2.4.2.)		
Workmanship	Brush not well made; head not evenly trimmed; wood component not smoothly finished; splintered or split grain; wood component not finished with varnish, lacquer, or paint	X	
	Brush not clean; evidence of dirt, or other foreign matter		X

4.2.2.4.2 *For dimensional compliance.* The sample unit for this examination shall be one brush. Examination of end items shall be made to determine compliance with di-

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mensional requirements. Any dimension not within the specified requirements shall be classified as a defect.

4.3 **Test.** The sample unit for test shall be one type F brush and the lot size shall be expressed in units of brushes. Sample brushes, examined as specified in 4.2.2.4, shall be tested as specified in 4.3.1. The requirements shall be applicable to the individual unit and the results reported as pass or fail. The inspection level shall be S-2; the AQL, 4.0.

4.3.1 *Bristle set (type F only).* Individual strands of bristle shall be held between the thumb and forefinger and pulled to determine compliance with 3.4.4.

4.4 **Certification.** The contractor shall submit to the contracting officer a certificate of compliance that the brushes he proposes to furnish meet the requirements of 3.3.1, 3.3.2 (for plastic), 3.4.1, and 3.4.2 (for plastic).

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A.* Unless otherwise specified, brushes of one type and size only, shall be packaged in a box conforming to PPP-B-566, style II, type A or B, class a; or to PPP-B-665, style optional; or to PPP-B-676, style optional. The gross weight of each box shall not exceed the weight limitation of the applicable box specification. Each box shall be waterproofed in accordance with MIL-P-116, method IC-2.

5.1.2 *Level B.* Twelve brushes, of one type and size only, shall be packaged in a box conforming to PPP-B-566; PPP-B-665; or PPP-B-676; styles and types optional. Each box shall be secured to prevent accidental opening.

5.1.3 *Level C.* Brushes shall be packaged in accordance with normal commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Unless otherwise specified, brushes, of one type and size, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or to PPP-B-636, class 2, V3c, style optional. Each box shall be waterproofed with a case liner conforming to and sealed in accordance with MIL-L-10547, except that fiberboard boxes may be waterproofed with tape in accordance with the appendix to the fiberboard box specification. The gross weight of each wood box shall not exceed 75 pounds; of each fiberboard box, the weight limitation of the box specification. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 *Level B.* Sixty boxes of type CR or type F brushes, 1 inch wide; or twelve boxes of type F brushes 3 inches wide, packaged as specified in 5.1, shall be packed in a fiberboard box conforming to PPP-B-636, type, class, and style optional. Bursting strength of the fiberboard shall be not less than 200 p.s.i. Closure shall be in accordance with the appendix to PPP-B-636.

5.2.3 *Level C.* Brushes shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Standard pack (civil agencies). The standard pack (for civil agencies) for the items covered by this specification shall be as specified in 5.1.2 and 5.2.2, interior and exterior containers, respectively.

5.4 Marking.

5.4.1 *Civil agencies.* In addition to any special marking required in the contract or or-

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der, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123, including specification classification, such as type and size. Unless otherwise specified (see 6.1), the cube shall be included in marking of shipping containers.

5.4.2 *Military agencies.* In addition to any special marking required in the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein (see 5.1.1, 5.2.1, and 5.4.1) and procurement documents should specify the following (see 4.4):

(*a*) Title, number, and date of this specification.
(*b*) Type, and size (type F) brush, required (see 1.2.1).
(*c*) Whether pig or hog bristle is required in type F brushes (see 3.4.1).
(*d*) Data, as required, regarding examination and tests; and inspection responsibility if other than as specified (see 4.1).
(*e*) Level of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.4).

6.1.1 *Standard pack (for civil agencies).* The standard pack requirements specified for level B, section 5 (Preparation for Delivery), are intended for use in procurement of Stores Stock replenishments. Procuring officers should use the standard pack provisions when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue, and/or shipment to eventual user. (See 5.3.)

6.2 Figures 1 and 2. The illustrations shown hereinafter as figures 1 and 2 are for the purpose of identification and are not intended to preclude any brush which otherwise meets the requirements of this specification.

6.3 Transportation data. Transportation descriptions and minimum weights applicable to the commodities covered herein are:

Rail:

Brushes, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Brushes, not otherwise indexed.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

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(See paragraph 3.2.)



FIGURE 1.—Type CR brush; with collar and rim



FIGURE 2.—Type F brush; flat, without collar

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